

YUNHAI XIANG

June 9, 2026

PERSONAL DETAILS

Name	Yunhai Xiang (in Chinese: 项云海, pronounced “yee-o-en-hi si-ang”)
Pronouns	He/Him/His/Himself
Email	yxiang72@uwo.ca (for encrypted emails: PGP Public Key)
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EDUCATION

Western University <i>Ph.D.</i> in Mathematics (in progress) Advisor: Chris Hall Specialization: Number Theory, Arithmetic Geometry, AI for Math	2025 — <i>Now</i>
Western University <i>M.Sc.</i> in Mathematics Advisor: Chris Hall	2024
University of Waterloo <i>B.Math.</i> in Pure Mathematics, Honors, with Distinction	2018 — 2023

SKILLS

Colored by level of mastery: *Expert*, *Advanced*, *Proficient*, *Familiar*, *Beginner*

Languages	Mandarin , English , French
Programming	Python , Lean , C/C++ , C#, Rust
Computing	SageMath , NumPy , SciPy , Pandas , PyTorch , XGBoost , Sklearn , Slurm
DSLs	LaTeX , SQL , HTML , CSS , GDScript
Softwares	Cinema 4D , Blender, Unity, Godot, Bevy
Hobbies	Magic Tricks

PUBLICATIONS

QEDBench: Quantifying the Alignment Gap in Automated Evaluation of University-Level Mathematical Proofs
Joint with Santiago Gonzalez, Alireza Amiri Bavandpour, Peter Ye, Edward Zhang, et al.
International Conference on Machine Learning (ICML) 2026

AWARDS

Mathematics Undergraduate Research Award	2022
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- Highly competitive undergrad research award of value \$10000 at University of Waterloo. Worked on research project in summer related to Weyl groups, supervised by Prof. Matthew Satriano.

ACADEMIC MEETINGS & PRESENTATIONS

Conferences Attendance & Presentations
Attended academic conferences and given research or expository talks

- FrontierMath Open Problems Workshop, Summer 2026, UToronto
- Homotopy Theory, K-theory, and Topological Data Analysis Conference, Summer 2026, UWO

- Using AI to Find Examples Workshop, under thematic program Universal Statistics in Number Theory, Winter 2026, Centre de Recherches Mathématiques, Université de Montréal
- University of Western Ontario Research in Computer Science (UWORCS), Winter 2026, UWO
- *Rigid Analytic Geometry*, Ontario Graduate Mathematics Conference, Summer 2024, UWaterloo
- *Cohomology of Grassmannians*, Canadian Undergraduate Mathematics Conference, Summer 2022, Université Laval

Seminar Attendance & Presentations

Attended academic seminars and given research or expository talks

- *Hyperelliptic Jacobians and Hasse–Witt Matrix*, Algebraic Geometry Student Seminar, UWO
- *Trace of Frobenius*, Winter 2026, Graduate Student Seminar, UWO
- *Brauer Groups of Fields Part II*, Winter 2026, Algebraic Geometry Student Seminar, UWO
- *Brauer Groups of Fields Part I*, Winter 2026, Algebraic Geometry Student Seminar, UWO
- *Weyl Group and Root Datum*, Summer 2025, Algebraic Groups Seminar, UWO
- *Hecke Operators*, Summer 2025, Algebraic Groups Seminar, UWO
- *Parabolic and Borel Subgroups*, Summer 2025, Algebraic Groups Seminar, UWO
- *Lie Algebras of Algebraic Groups*, Summer 2025, Algebraic Groups Seminar, UWO
- *Non-Archimedean Fields*, Summer 2024, Rigid Analytic Geometry Seminar, UWO
- *Admissible Topology*, Summer 2024, Rigid Analytic Geometry Seminar, UWO
- *Cones and Affine Toric Varieties*, Summer 2024, Toric Varieties Learning Seminar, UWO
- *Galois Cohomology*, Winter 2024, Graduate Student Seminar, UWO
- *Derived Categories*, Fall 2022, Algebraic Geometry Working Seminar, UWaterloo

TEACHING & MENTORING

Course Instructorship

- Coming soon.

Teaching Assistantship

Proctored and graded exams, graded homeworks, tutored at Help Centre, and taught tutorial sessions

- MATH 3150, *Elementary Number Theory I*, Fall 2024
- MATH 2155, *Mathematical Structures*, Fall 2025
- CALCULUS 1501, *Calculus II for Mathematical and Physical Sciences*, Winter 2025, Winter 2026, Summer 2026
- MATH 1229, *Methods of Matrix Algebra*, Summer 2024, Summer 2025
- MATH 1228, *Methods of Finite Mathematics*, Summer 2024, Summer 2025
- CALCULUS 1000, *Calculus I*, Winter 2024, Summer 2024, Winter 2025, Fall 2025, Winter 2026

Mathematical Outreach Lectures

Gave expository lectures to high school students or undergraduate students on advanced mathematical topics

- *Surreal Numbers and the Game of Go*, Summer 2025, Western Summer Math Camp (Grade 10)
- *Mathematical Magic Tricks: Gilbreath's Principle*, Summer 2024, Western Summer Math Camp (Grade 11)
- *Why is There no Quintic Formula*, Summer 2024, Western Summer Math Camp (Grade 10)
- *Category Theory Demystified*, Winter 2023, Short Attention Span Math Seminar (Undergraduate)

Directed Reading Program Mentorship

Mentored undergraduate students on reading projects, culminating in a presentation

- Jude McNorgan Lubos, *Introduction to Quadratic Reciprocity*, Winter 2026

SERVICE & VOLUNTEER WORK

Seminar Organization

Research or learning seminars organized or co-organized

- Algebraic Geometry Student Seminar, Summer 2026
- Algebraic Geometry Student Seminar, Winter 2026, co-organized with Alex Zwart
- Algebraic Geometry Student Seminar, Fall 2024, co-organized with Alex Zwart
- Toric Varieties Learning Seminar, Summer 2024, co-organized with Nikolas Rieke